

Topological Integrity Gravity:
A Structural Constraint on Horizon Formation in Quadratic $f(R)$
Gravity

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(Dated: May 5, 2026)

Abstract

We formulate Topological Integrity Gravity (TIG) as a structural constraint on horizon formation within a quadratic $f(R)$ framework. The central idea is that horizon formation is governed by a structural consistency condition rather than being an arbitrary feature of a given solution.

In the static, spherically symmetric sector, this condition reduces to a cubic normal form exhibiting a saddle-node bifurcation. This introduces a critical parameter separating horizon-bearing and horizonless configurations.

The Schwarzschild solution is recovered in the limit of vanishing correction scale. Observable consequences include modifications of the photon sphere and near-critical scaling of echo delays. These effects are discussed as possible diagnostic channels, not as unique observational signatures at the present level of modeling.

I. INTRODUCTION

General Relativity provides a highly successful description of gravitational phenomena across a wide range of scales [1, 2]. Nevertheless, when considering extensions of General Relativity or regularized geometries, the structure of horizons is no longer uniquely fixed.

Modified gravity theories such as $f(R)$ gravity introduce additional curvature terms and scalar degrees of freedom, enlarging the space of admissible solutions [3, 4]. This raises the question of whether additional structural principles are required to identify physically consistent configurations.

Regular black hole models provide explicit examples where singularities are removed while horizon-like structures remain present [5, 6]. However, such constructions are typically model-dependent and do not by themselves define a universal criterion for horizon formation.

In this work, we propose that horizon formation is governed by a structural consistency condition. Rather than modifying the field equations directly, TIG restricts the admissible form of the horizon equation. The resulting framework is therefore best understood as defining a constrained solution sector rather than a complete replacement for the gravitational field equations.

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II. STRUCTURAL HORIZON EQUATION

We consider the dimensionless equation

$$x^3 - x^2 + \beta^3 = 0, \quad (1)$$

with

$$x = \frac{r_H}{2M}, \quad \beta = \frac{r_c}{2M}. \quad (2)$$

Here r_H is the horizon radius, M is the mass parameter, and r_c is the internal correction scale. The dimensionless ratio β compares the correction scale to the Schwarzschild radius.

Equation (1) is not introduced as an arbitrary fitting ansatz. It is interpreted as the minimal algebraic structure capable of describing a change in the number of horizon solutions under smooth variation of a control parameter.

From the perspective of bifurcation theory, this is the normal form of a saddle-node transition. It captures the merger of two real branches and their subsequent disappearance beyond a critical value of the control parameter.

III. QUADRATIC $f(R)$ FRAMEWORK

We consider the quadratic action

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi} (R - 2\Lambda + \alpha R^2) \right] + S_{\text{matter}}. \quad (3)$$

The quadratic term introduces higher-curvature corrections and, in the standard interpretation of $f(R)$ gravity, an additional scalar degree of freedom. In the present paper, however, the full dynamical solution space of the theory is not constructed. Instead, we use a reduced static sector to analyze a structural constraint on horizon formation.

The role of the quadratic $f(R)$ framework is therefore to provide a conservative setting in which higher-curvature corrections can be discussed while preserving the Schwarzschild limit. TIG is not assumed to replace the underlying field equations at this stage.

IV. EFFECTIVE GEOMETRY

A minimal static and spherically symmetric realization consistent with the structural constraint is given by

$$ds^2 = -F(r)dt^2 + \frac{dr^2}{F(r)} + r^2 d\Omega^2, \quad (4)$$

with

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}. \quad (5)$$

Equivalently, this can be written in terms of an effective mass profile

$$m(r) = \frac{Mr^3}{r^3 + r_c^3}, \quad (6)$$

so that

$$F(r) = 1 - \frac{2m(r)}{r}. \quad (7)$$

The form (5) satisfies three requirements. First, it reproduces the Schwarzschild lapse function in the limit $r_c \rightarrow 0$. Second, it regularizes the central region by replacing the singular mass profile with a smooth effective profile. Third, the horizon condition reduces directly to the cubic structural equation (1).

A. Reduction of the Horizon Condition

The horizon condition $F(r_H) = 0$ implies

$$1 - \frac{2Mr_H^2}{r_H^3 + r_c^3} = 0. \quad (8)$$

Multiplying by $r_H^3 + r_c^3$ gives

$$r_H^3 + r_c^3 - 2Mr_H^2 = 0. \quad (9)$$

Introducing the dimensionless variables

$$x = \frac{r_H}{2M}, \quad \beta = \frac{r_c}{2M}, \quad (10)$$

we obtain

$$(2M)^3 x^3 + (2M)^3 \beta^3 - 2M(2M)^2 x^2 = 0. \quad (11)$$

Dividing by $(2M)^3$ yields

$$x^3 - x^2 + \beta^3 = 0, \quad (12)$$

which coincides with Eq. (1).

V. INTERPRETATION OF THE SCALE r_c

The parameter r_c denotes the characteristic scale at which the structural correction becomes relevant. It should not be interpreted as an independently postulated horizon radius. Rather, it controls the departure from the classical Schwarzschild geometry in the near-core region.

In the quadratic $f(R)$ setting, r_c may be interpreted as an effective length scale associated with higher-curvature corrections. A complete derivation of this scale from the covariant field equations would require solving the modified equations with appropriate boundary and regularity conditions. That derivation is beyond the minimal construction considered here.

For the present analysis, only the dimensionless ratio

$$\beta = \frac{r_c}{2M} \quad (13)$$

is required. This parameter controls the horizon transition and determines the approach to the critical point.

VI. CRITICAL HORIZON STRUCTURE

The discriminant of Eq. (1) is

$$\Delta(\beta) = \beta^3(4 - 27\beta^3). \quad (14)$$

Criticality occurs when two roots merge. Equivalently, the critical point is determined by

$$f(x_c, \beta_c) = 0, \quad \partial_x f(x_c, \beta_c) = 0, \quad (15)$$

where

$$f(x, \beta) = x^3 - x^2 + \beta^3. \quad (16)$$

Since

$$\partial_x f = 3x^2 - 2x = x(3x - 2), \quad (17)$$

the nontrivial stationary point is

$$x_c = \frac{2}{3}. \quad (18)$$

Substitution into $f(x_c, \beta_c) = 0$ gives

$$\beta_c = \left(\frac{4}{27}\right)^{1/3}. \quad (19)$$

At this point, two horizon branches merge, producing a saddle-node transition. For $\beta < \beta_c$, the effective geometry contains two positive horizon branches. At $\beta = \beta_c$, the horizons coincide. For $\beta > \beta_c$, the physical horizon branch is absent.

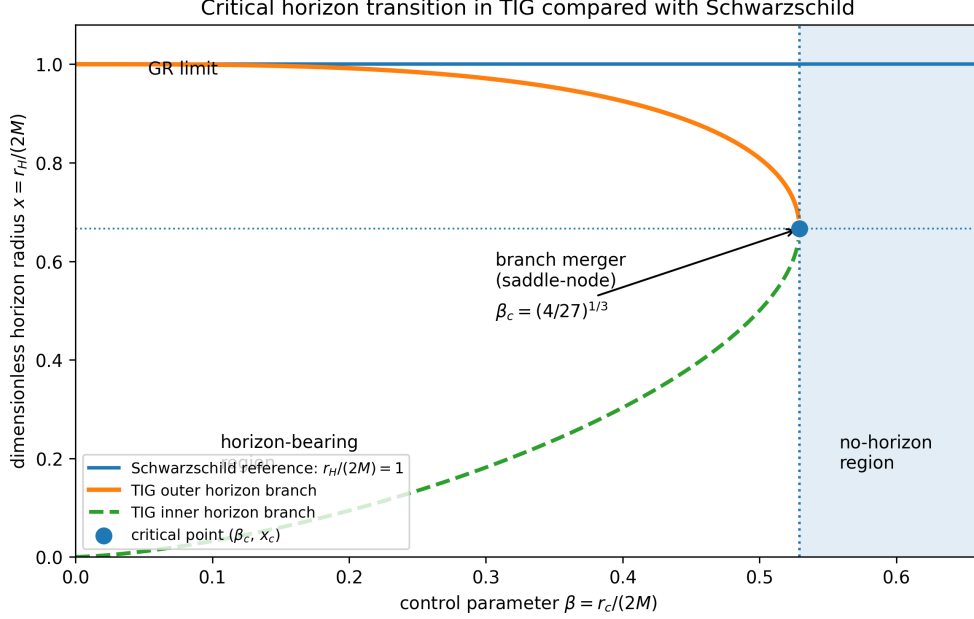


FIG. 1. Horizon branches as a function of the dimensionless control parameter β . The branches merge at the critical point (β_c, x_c) .

VII. NEAR-CRITICAL SCALING

To analyze the behavior near criticality, write

$$x = x_c + \delta x, \quad \beta = \beta_c - \delta\beta, \quad (20)$$

with $\delta\beta > 0$.

A Taylor expansion around (x_c, β_c) gives, to leading order,

$$0 \approx \frac{1}{2} f_{xx}(x_c, \beta_c) (\delta x)^2 - f_\beta(x_c, \beta_c) \delta\beta. \quad (21)$$

Using

$$f_{xx}(x_c, \beta_c) = 2, \quad f_\beta(x_c, \beta_c) = 3\beta_c^2, \quad (22)$$

one obtains

$$(\delta x)^2 \approx 3\beta_c^2 \delta\beta. \quad (23)$$

Therefore,

$$x_{\pm} - x_c \approx \pm \sqrt{3} \beta_c (\beta_c - \beta)^{1/2}. \quad (24)$$

The square-root scaling is the universal signature of a saddle-node bifurcation. It is independent of the detailed microscopic interpretation of the effective geometry.

VIII. PHOTON SPHERE STRUCTURE

The photon sphere is determined by the condition for unstable circular null geodesics,

$$rF'(r) - 2F(r) = 0. \quad (25)$$

In the Schwarzschild limit, this gives the standard value

$$r_{\text{ph}}^{\text{GR}} = 3M. \quad (26)$$

For finite r_c , Eq. (25) is solved numerically. We define the relative deviation

$$\Delta_{\text{ph}} = \frac{r_{\text{ph}}^{\text{TIG}} - 3M}{3M}. \quad (27)$$

For small β , the leading correction scales as

$$\Delta_{\text{ph}} \sim -\frac{16}{27}\beta^3. \quad (28)$$

This provides a direct observable channel because the photon sphere controls the shadow and photon-ring structure of compact objects.

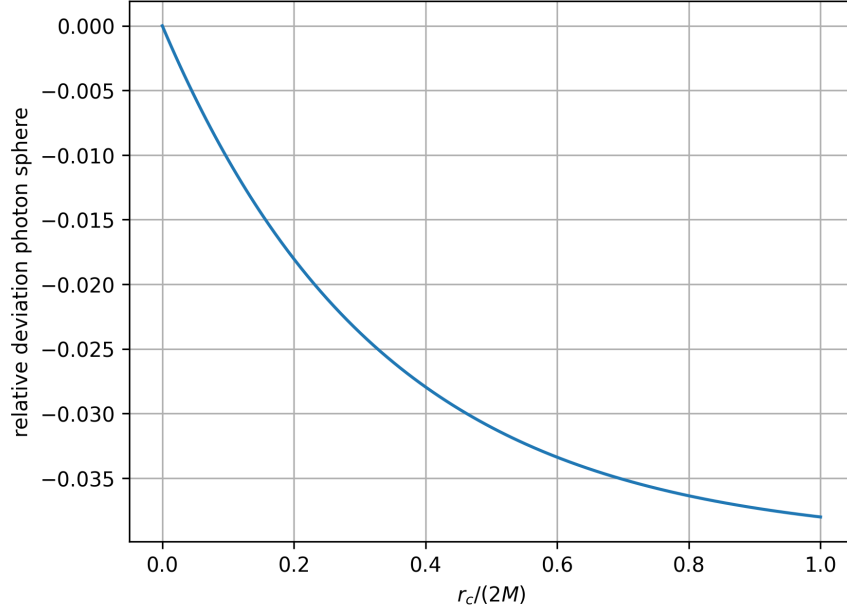


FIG. 2. Relative deviation of the photon sphere radius from the Schwarzschild value.

IX. ECHO PHYSICS

A schematic estimate of the echo delay is

$$\Delta t_{\text{echo}} \sim \int \frac{dr}{F(r)}. \quad (29)$$

Near the critical point, the lapse function develops an increasingly shallow minimum close to the degenerate horizon. This enhances propagation times in the near-horizon region.

Writing the near-critical behavior schematically as

$$F(r) \sim (r - r_c)^2 + \epsilon, \quad \epsilon \sim \beta_c - \beta, \quad (30)$$

one obtains

$$\Delta t_{\text{echo}} \sim (\beta_c - \beta)^{-1/2}. \quad (31)$$

This scaling should be interpreted as a local near-critical estimate, not as a complete waveform model.

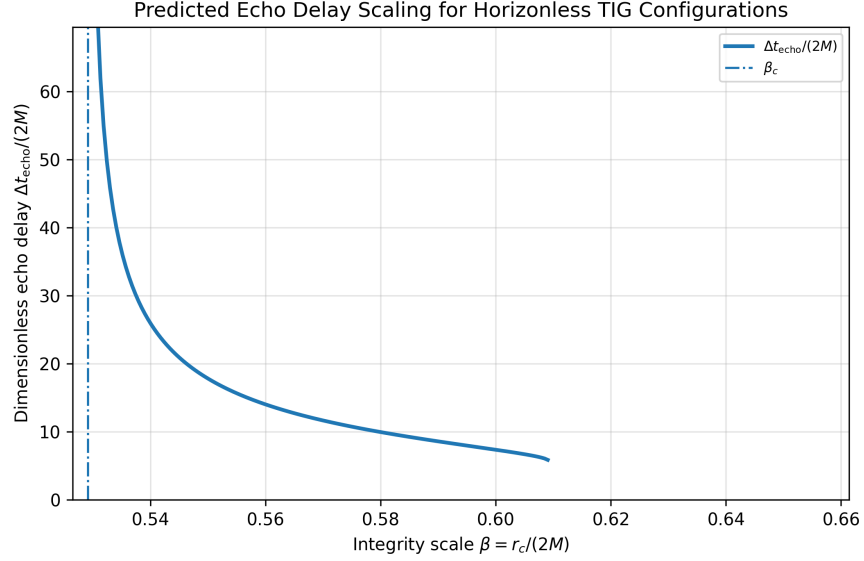


FIG. 3. Estimated echo-delay scaling near the critical point.

X. OBSERVATIONAL INTERFACE

The leading photon-sphere correction suggests a corresponding shift in the angular shadow size,

$$\frac{\delta R_{\text{sh}}}{R_{\text{sh}}^{\text{GR}}} \sim -\frac{16}{27}\beta^3. \quad (32)$$

This relation should be interpreted as a leading-order diagnostic rather than as a complete image-domain prediction. A realistic comparison with observations requires marginalization over spin, inclination, accretion structure, and plasma effects.

Similarly, echo-delay scaling provides a possible gravitational-wave diagnostic,

$$\Delta t_{\text{echo}} \sim (\beta_c - \beta)^{-1/2}. \quad (33)$$

The absence of echoes can therefore be used to constrain how close a system lies to the critical regime, although such constraints are strongly model-dependent.

XI. INFERENCE FRAMEWORK

A minimal likelihood based on shadow-size measurements may be written as

$$\theta_{\text{TIG}} = \theta_{\text{GR}} \left(1 - \frac{16}{27}\beta^3 \right), \quad (34)$$

and

$$\mathcal{L}_{\text{EHT}} \propto \exp \left[-\frac{(\theta_{\text{obs}} - \theta_{\text{TIG}})^2}{2\sigma^2} \right]. \quad (35)$$

A combined phenomenological likelihood can include gravitational-wave constraints,

$$\mathcal{L}_{\text{tot}} = \mathcal{L}_{\text{EHT}} \mathcal{L}_{\text{GW}}. \quad (36)$$

This likelihood is intended only as an illustrative inference pipeline. It is not a replacement for a full astrophysical analysis.

XII. COVARIANT INTERPRETATION

A covariant interpretation can be formulated in terms of null expansions. Let ℓ^μ and n^μ be future-directed null vectors normalized by

$$\ell^\mu n_\mu = -1. \quad (37)$$

The outgoing null expansion is

$$\theta_+ = q^{\mu\nu} \nabla_\mu \ell_\nu, \quad (38)$$

where $q_{\mu\nu}$ is the induced metric on the transverse two-surface.

A marginal surface satisfies

$$\theta_+ = 0. \quad (39)$$

The TIG critical sector corresponds to a degeneracy of the marginal-surface condition,

$$\theta_+ = 0, \quad \mathcal{L}_n \theta_+ = 0. \quad (40)$$

In the static sector, this reduces to the double-root condition

$$f(x_c, \beta_c) = 0, \quad \partial_x f(x_c, \beta_c) = 0. \quad (41)$$

XIII. VARIATIONAL EMBEDDING AND CONSTRAINT CONSISTENCY

A schematic variational embedding is

$$S_{\text{eff}} = S_{f(R)} + \int d^4x \sqrt{-g} \lambda(x) \theta_+ \mathcal{L}_n \theta_+. \quad (42)$$

The constrained field equations take the formal form

$$E_{\mu\nu}^{f(R)} + \lambda \mathcal{K}_{\mu\nu} = 8\pi T_{\mu\nu}. \quad (43)$$

Consistency with covariant conservation requires

$$\nabla^\mu (\lambda \mathcal{K}_{\mu\nu}) = 0, \quad (44)$$

assuming

$$\nabla^\mu T_{\mu\nu} = 0. \quad (45)$$

At the present level, $\lambda(x)$ acts as a Lagrange multiplier enforcing the structural constraint rather than as a new propagating matter field.

The propagating degrees of freedom are therefore inherited from quadratic $f(R)$ gravity: the two tensor modes of GR and the scalar mode associated with the curvature-squared correction. A complete Hamiltonian analysis would be needed to verify this statement beyond the reduced sector.

XIV. NUMERICAL PIPELINE

A reproducible numerical analysis proceeds as follows. First, for each value of β , one solves

$$x^3 - x^2 + \beta^3 = 0 \quad (46)$$

and selects the positive real roots. The outer horizon is identified as the largest positive root.

Second, the photon-sphere condition

$$x \frac{dF}{dx} - 2F = 0 \quad (47)$$

is solved numerically. The relative deviation is then computed from

$$\Delta_{\text{ph}} = \frac{2x_{\text{ph}} - 3}{3}. \quad (48)$$

Third, a regulated echo-delay proxy may be evaluated by integrating $1/F$ between the relevant near-horizon turning points, with a cutoff avoiding coordinate singularities.

These steps reproduce the horizon, photon-sphere, and echo-delay plots used in the present work.

XV. CONCLUSION

We have introduced TIG as a structural constraint on horizon formation within a quadratic $f(R)$ setting. In the static sector, the horizon condition reduces to a cubic normal form exhibiting a saddle-node bifurcation.

The Schwarzschild solution is recovered in the limit $r_c \rightarrow 0$, while finite r_c introduces a critical transition separating horizon-bearing and horizonless configurations.

The framework provides possible observational interfaces through photon-sphere deviations and echo-delay scaling. These signatures are testable but not unique, and a full observational analysis requires realistic astrophysical modeling.

The present formulation should therefore be understood as a constrained solution sector rather than a complete dynamical theory. Its main result is the identification of a structurally stable mechanism for horizon transitions.

Appendix A: Illustrative MCMC Inference Pipeline

The following minimal inference pipeline illustrates how the TIG control parameter β may be constrained from shadow-size measurements. It is intended as a reproducibility example and not as a substitute for a full astrophysical analysis.

```
import numpy as np
import emcee
import matplotlib.pyplot as plt

beta_c = (4.0 / 27.0)**(1.0 / 3.0)

def delta_shadow(beta):
    return -(16.0 / 27.0) * beta**3

def theta_tig(theta_gr, beta):
    return theta_gr * (1.0 + delta_shadow(beta))

def echo_amplitude(beta, A0=0.05):
```

```

    eps = beta_c - beta
    if eps <= 0:
        return np.inf
    return A0 / np.sqrt(eps)

data_eht = [
    {"name": "M87*", "theta_obs": 42.0, "theta_gr": 42.0, "sigma"
     : 3.0},
    {"name": "Sgr_A*", "theta_obs": 51.8, "theta_gr": 51.8, "sigma"
     : 2.3},
]

def log_prior(beta):
    if 0.0 <= beta < beta_c:
        return 0.0
    return -np.inf

def log_likelihood_eht(beta):
    logL = 0.0
    for d in data_eht:
        model = theta_tig(d["theta_gr"], beta)
        resid = d["theta_obs"] - model
        logL += -0.5 * (resid / d["sigma"])**2
    return logL

def log_likelihood_echo(beta, sigma_A=0.5):
    A = echo_amplitude(beta)
    if not np.isfinite(A):
        return -np.inf
    return -0.5 * (A / sigma_A)**2

```

```

def log_probability(theta):
    beta = theta[0]
    lp = log_prior(beta)
    if not np.isfinite(lp):
        return -np.inf
    return lp + log_likelihood_eht(beta) + log_likelihood_echo(beta
    )

ndim = 1
nwalkers = 32
nsteps = 5000
burnin = 1000

rng = np.random.default_rng(42)
initial = rng.uniform(0.001, 0.3, size=(nwalkers, ndim))

sampler = emcee.EnsembleSampler(nwalkers, ndim, log_probability)
sampler.run_mcmc(initial, nsteps, progress=True)

samples = sampler.get_chain(discard=burnin, flat=True)
beta_samples = samples[:, 0]

print("beta_c=", beta_c)
print("median_beta=", np.median(beta_samples))
print("68_percent_interval=", np.percentile(beta_samples, [16,
    84]))
print("95_percent_upper_bound=", np.percentile(beta_samples, 95))

plt.hist(beta_samples, bins=50, density=True)

```

```
plt.xlabel("beta")
plt.ylabel("posterior_density")
plt.tight_layout()
plt.savefig("tig_beta_posterior.png", dpi=200)
```

The numerical values used above are illustrative benchmark inputs. A full observational analysis would require image-domain modeling, marginalization over spin, inclination, plasma structure, accretion morphology, and gravitational-wave systematics.

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